DJ Mantra — Manual

DJ Mantra is a DJ app for Android phones, made for long, smooth blends: two decks, a mixer that only cuts, phrase lights that show beat 1 of every 8-beat phrase, and sections that show where the music changes. It runs on the open-source Mixxx 2.5 engine and works with the Hercules DJControl Mix Ultra controller over USB or Bluetooth.

This manual has two parts: **how to use the app**, then **the technology behind it**. Menu → MANUAL shows it in the app; EXPORT MD and EXPORT PDF save it to *Documents/DJ Mantra/Manual*, where it can be printed.

The pictures are drawn from the app's own screens (its interface files, with sample songs), at the size of a Pixel 7.

Part 1 — Using DJ Mantra

1. First start

1. Install the APK and open DJ Mantra.
2. Allow **All files access** when Android asks: the app reads your music and writes its documents.
3. Put your music anywhere under **Music** (or in *Music/djmantra_app*). The app analyses every song once: tempo (BPM), beat grid, key, sections and loudness.
4. Connect the controller (optional): Menu → **CONTROLLER: USB** or **CONTROLLER: BLUETOOTH**.

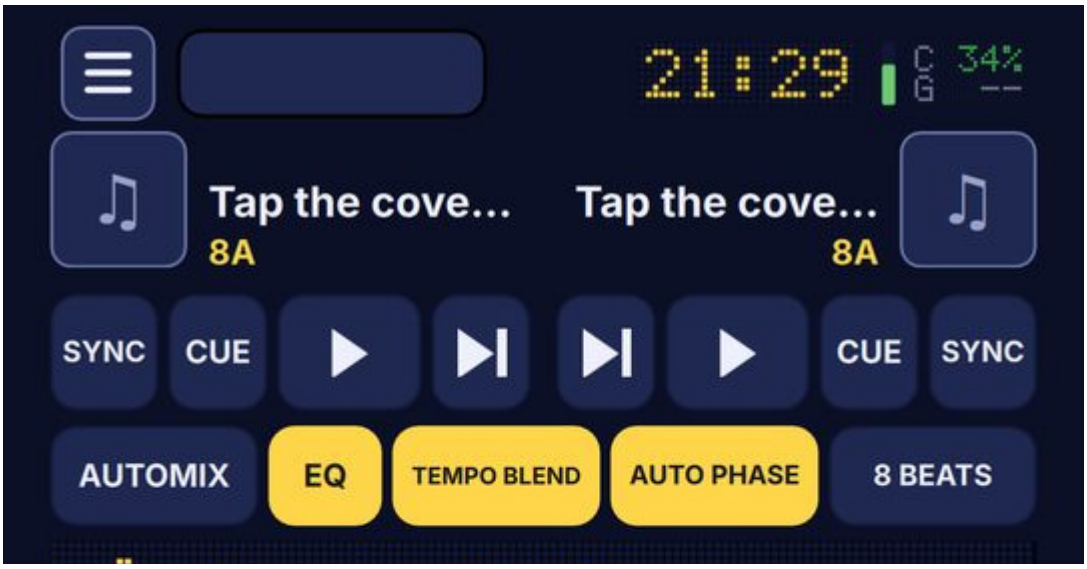
2. The screen

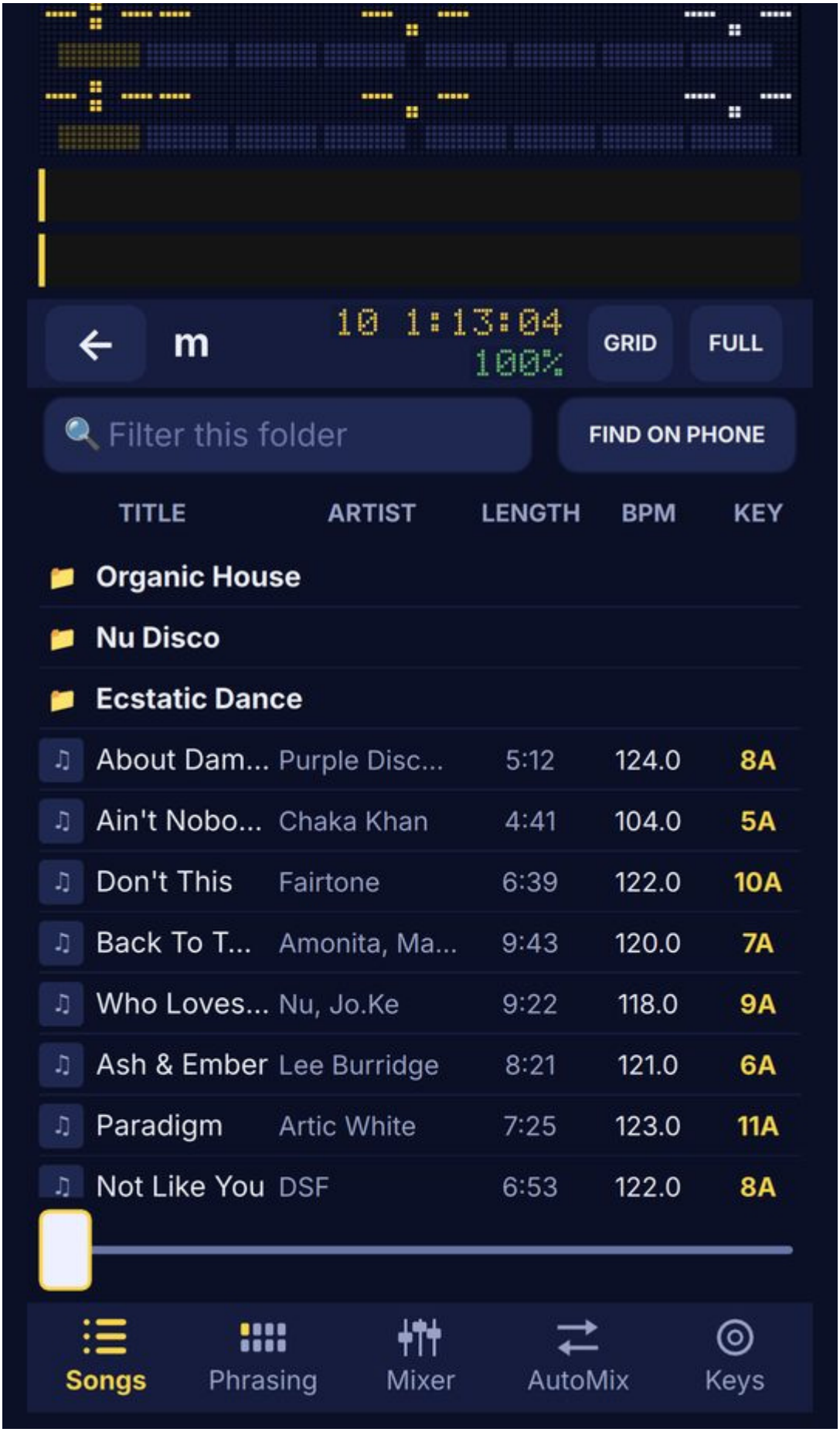
The top half is always the same, whatever tab is open:

- **Top bar**: the menu (☰), the two decks' keys and whether they match (e.g. *11B > 12B, +1 energy up*), the clock, the CPU / GPU load and the battery. Long press on the clock starts or stops recording (the clock turns red).
- **Deck headers**: cover, title, artist and key of each deck. Tap a cover to load the highlighted song into that deck.
- **Transport**: SYNC, CUE, PLAY / PAUSE and NEXT for each deck.
- **Modes**: AUTOMIX, EQ, TEMPO BLEND, AUTO PHASE and the beats selector (8 / 16 / 32).
- **The matrix display**: per deck the time left, the countdown to the next section and the bar.beat, with the running phrase lights.
- **Overview lanes**: each song's whole waveform, with its sections.
- **Crossfader** at the bottom, then the five tabs: **Songs, Phrasing, Mixer, AutoMix, Keys**. The controller's pad-mode buttons switch the same tabs.

Every change made with a finger shows a short message (a *toast*) above the controls, because the finger covers what it moves. Every long press is 1 second. Nothing snaps: resets and automatic moves glide.

3. Songs

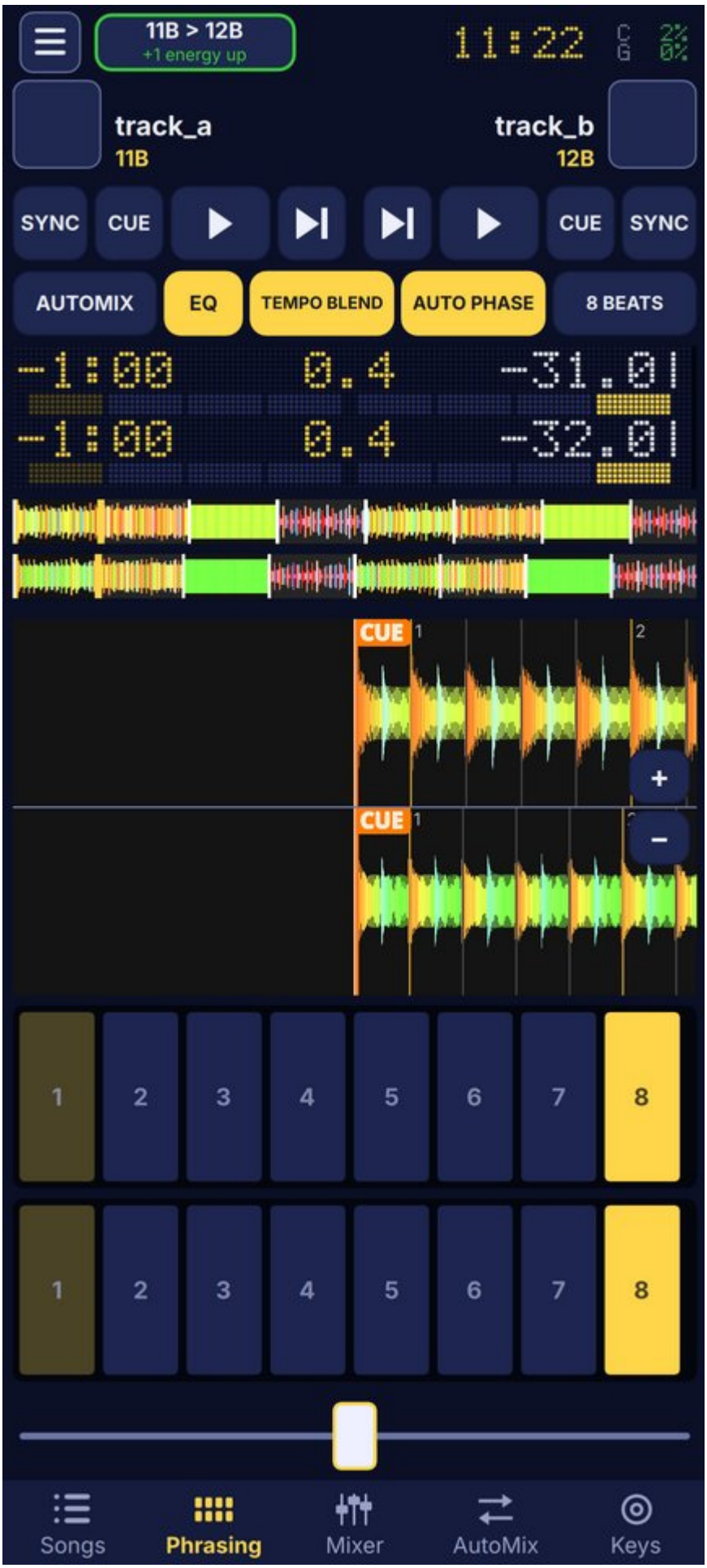




- Folders and songs, with **TITLE, ARTIST, LENGTH, BPM, KEY** columns; tap a column to sort by it.
- **Tap a song**: it loads into the free deck (the one not playing; when none plays, deck 1, deck 2, deck 1 ...). A playing deck is never replaced.
- **Long press a song**: add it to a playlist.

- **Find a song** filters this folder; **FIND ON PHONE** searches all your music.
- **Playlists** is a folder like the others; empty folders and playlists are hidden.
- **SUNO**: paste links to your Suno songs; **GET** opens Suno (log in once) and saves the song you download there to *Music/djmantra_app/Suno*.
- **TIDAL**: your TIDAL playlists (import into the app), send an app playlist to TIDAL, or SYNC ALL.
- **FULL** gives the list the whole screen.

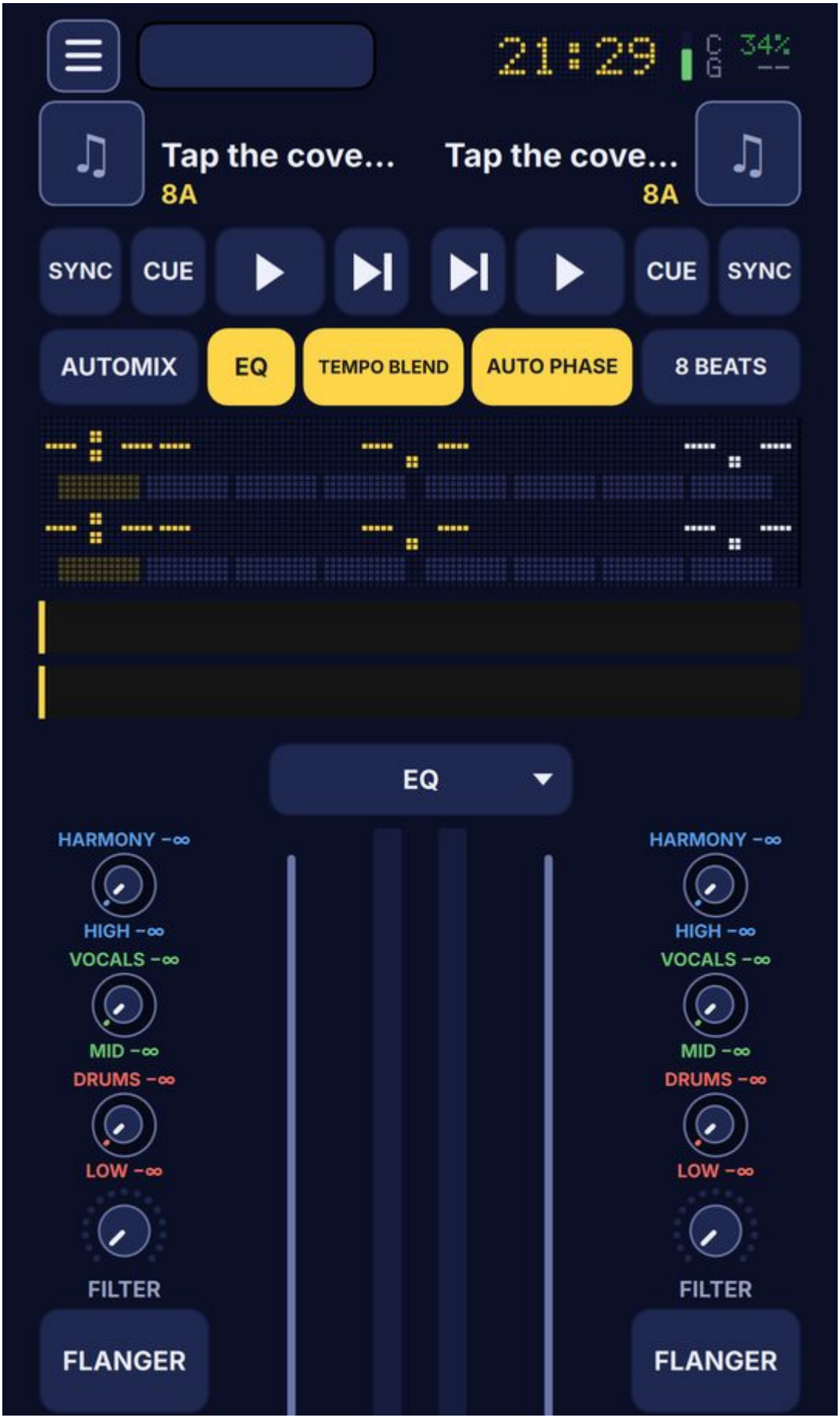
4. Phrasing

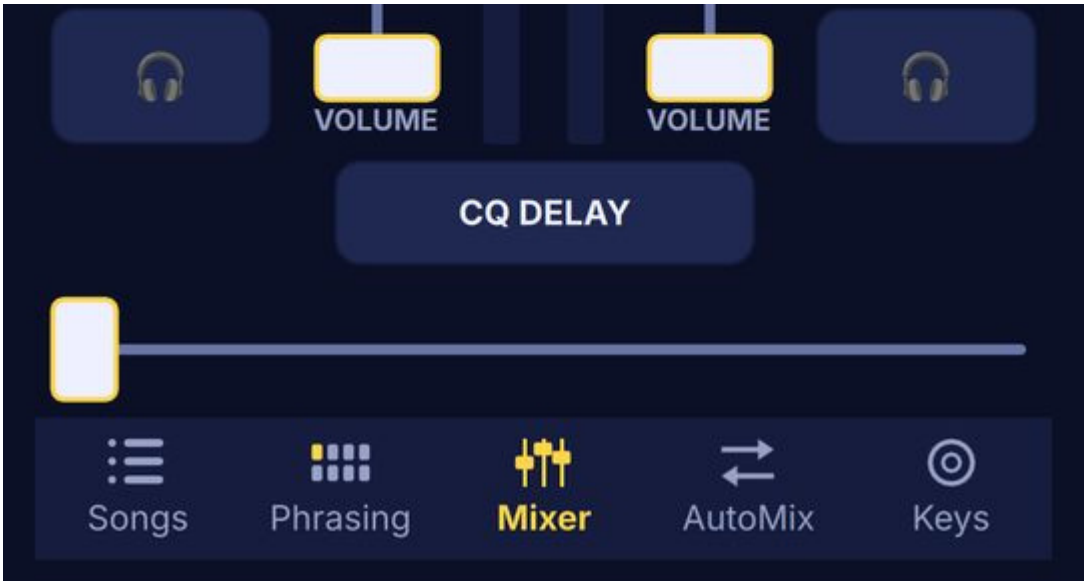


- The two decks' scrolling waveforms, one over the other, with the same zoom (**+ / –**), so the beats line up by eye. Bar lines are numbered.

- **Phrase pads 1–8** for each deck: the lit pad is the current beat of the 8-beat phrase. On a stopped deck, tap pad N to start it from its phrase beat N, in time with the other deck.
- Drag the waveform to move through the song.

5. Mixer





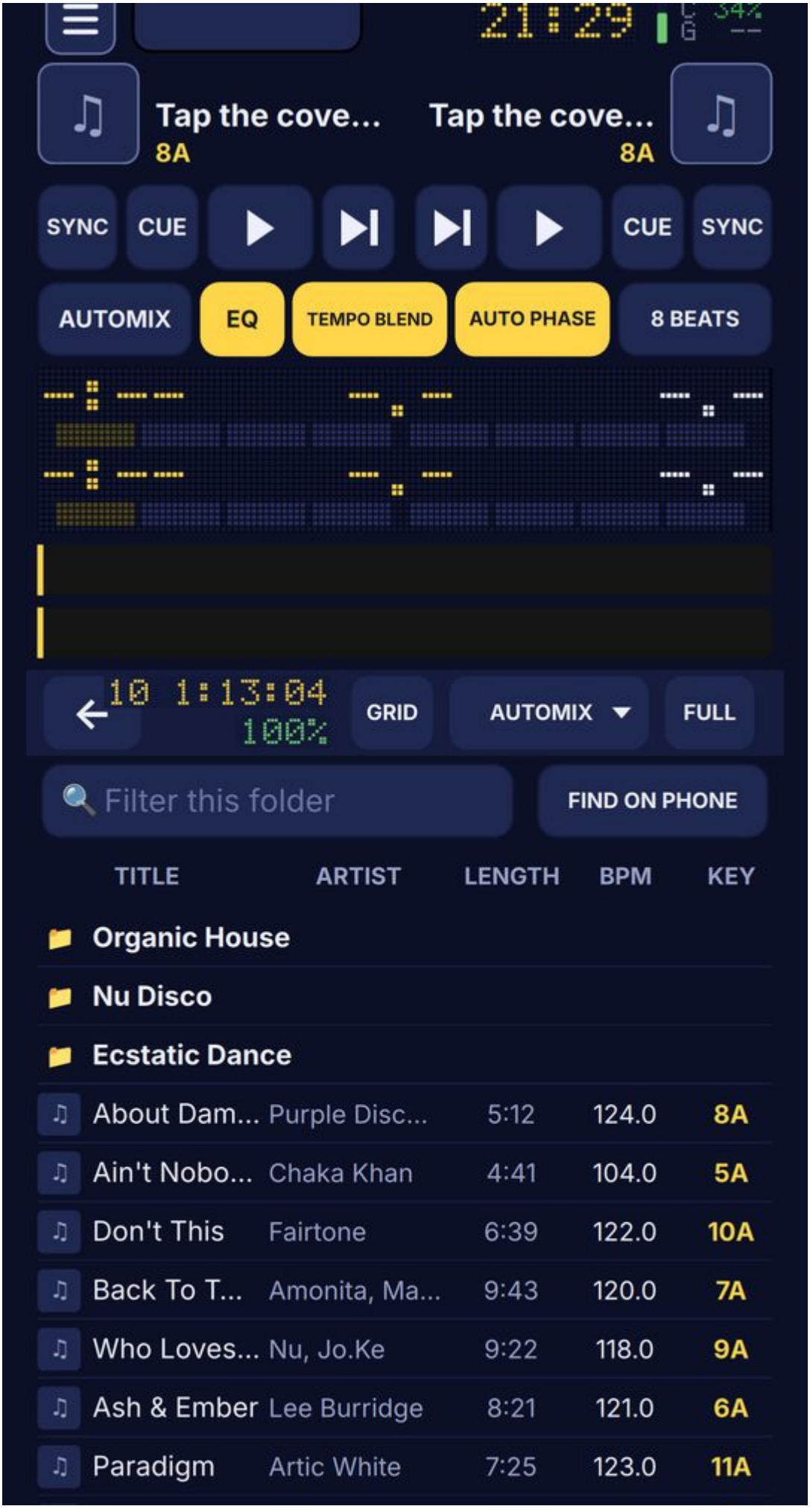
- **EQ / NEURAL MIX** at the top switches what the knobs do: the **Equalizer** (HIGH, MID, LOW) or **Neural Mix** (HARMONY, VOCALS, DRUMS).
- Each knob is **concentric**, like a big studio mixer's: the inner knob is the active mode, the outer ring the other mode. A finger turns the inner knob; the controller turns both (SHIFT for the outer). Both labels share one colour: HIGH / HARMONY blue, MID / VOCALS green, LOW / DRUMS red.
- **All EQs only cut**: the top is 0 dB, turning down cuts to a full kill. There is no boost. Long press resets to 0 dB.
- **FILTER** (bottom knob): a Pioneer DJM filter. Middle is off, left is a low-pass, right a high-pass. Long press resets it.
- **FX button**: tap = the deck's effect on / off; hold = choose the effect (FLANGER, UNDERWATER, COOLDOWN, ECHO OUT, DJM DELAY, DJM ECHO, DJM TRANS, DJM PAN) and its DEPTH.
- **Volume faders** and the **headphones** (🎧) buttons for each deck. Long press a fader to use it as the PITCH fader.
- Loading a new song resets its EQs and Neural Mix, so every song starts clean.

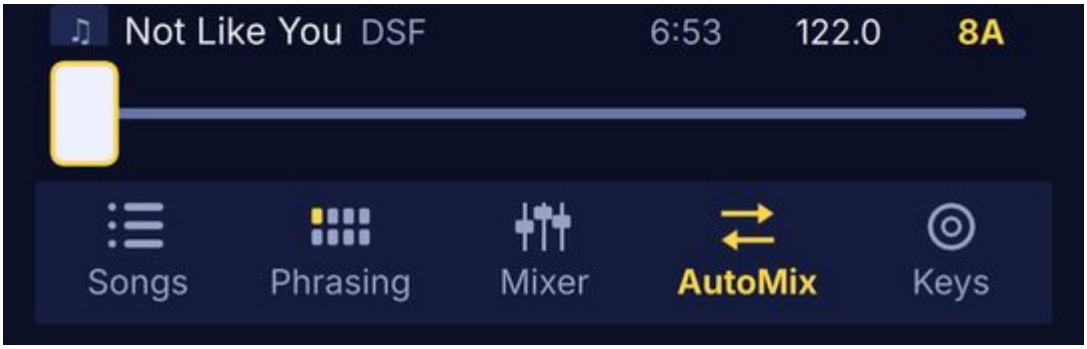


Ghosts. A knob on the controller does not make the screen's value jump: until the hardware knob reaches the value (or passes it), a **half-transparent ghost** shows where the hardware knob is. Then they move together. This works for the EQ knobs, FILTER and the effect depth.

6. AutoMix



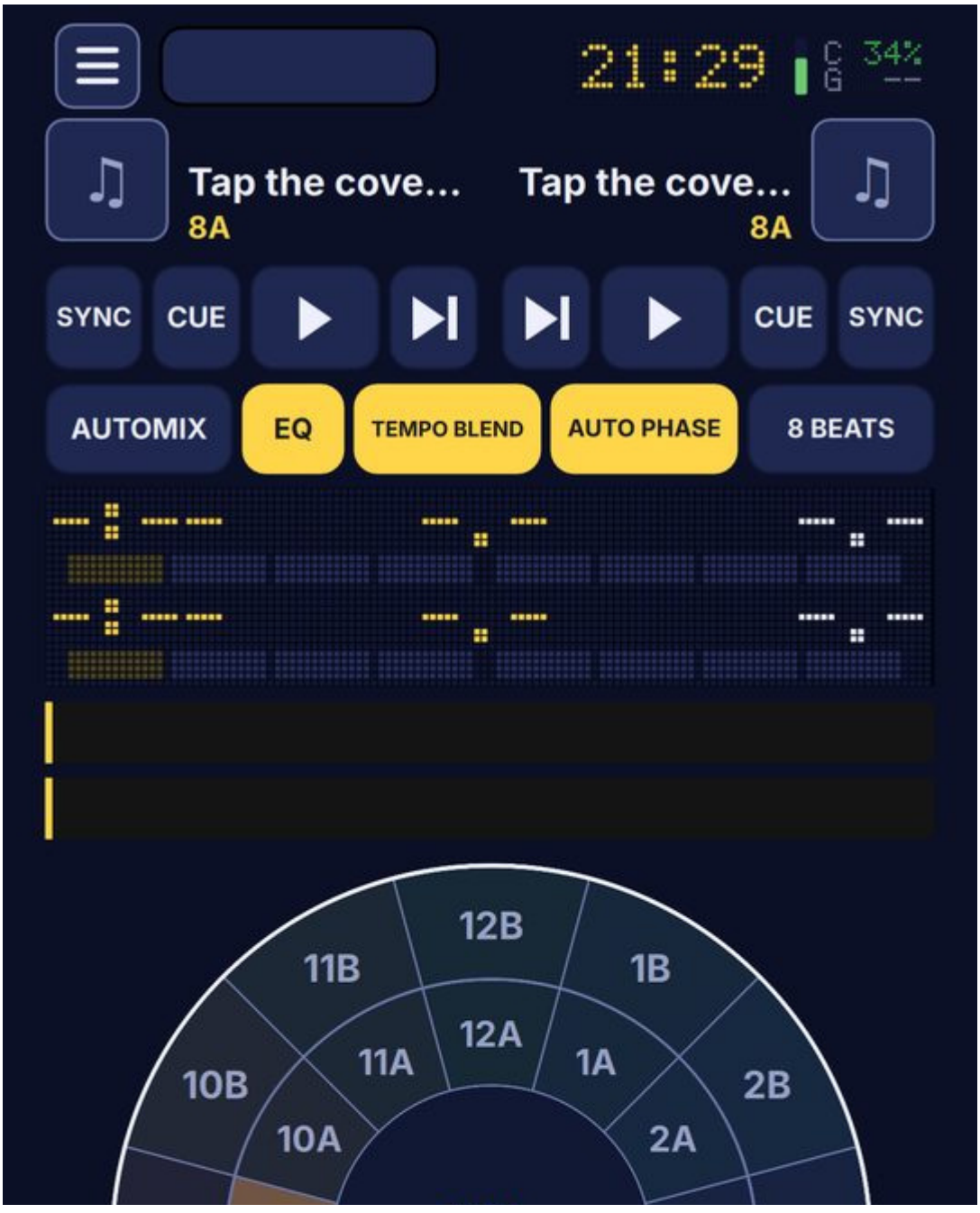

21:29
Tap the cove...
8A
Tap the cove...
8A
SYNC
CUE
CUE
SYNC
AUTOMIX
EQ
TEMPO BLEND
AUTO PHASE
8 BEATS
10 1:13:04
100%
GRID
AUTOMIX
FULL
Filter this folder
FIND ON PHONE
TITLE
ARTIST
LENGTH
BPM
KEY
Organic House
Nu Disco
Ecstatic Dance
About Dam... Purple Disc... 5:12 124.0 8A
Ain't Nobo... Chaka Khan 4:41 104.0 5A
Don't This Fairtone 6:39 122.0 10A
Back To T... Amonita, Ma... 9:43 120.0 7A
Who Loves... Nu, Jo.Ke 9:22 118.0 9A
Ash & Ember Lee Burridge 8:21 121.0 6A
Paradigm Artic White 7:25 123.0 11A

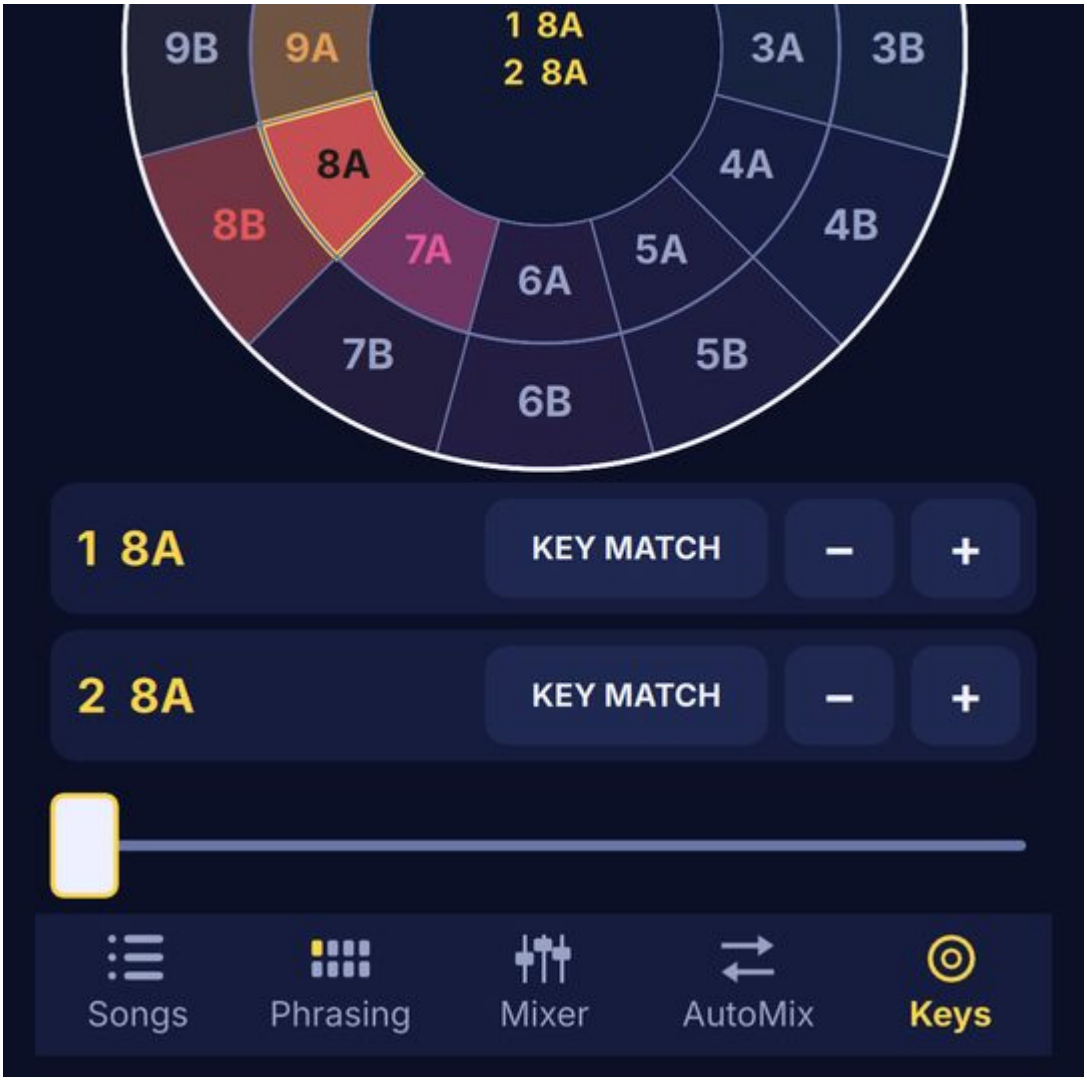


AutoMix plays the current folder or playlist by itself, song after song. Open the **AUTOMIX** menu to set it and START or STOP; PREVIOUS SONG / NEXT SONG choose what comes next. The **AUTOMIX** button above the matrix is a single action: it glides the crossfader to the other deck over the time set in Settings (in beats or seconds), while **EQ** and **TEMPO BLEND**, when lit, follow the crossfader:

- **EQ** swaps the bass first, then the highs, then the mids.
- **TEMPO BLEND** glides the tempo from the outgoing song's to the incoming song's.
- **AUTO PHASE** starts the incoming deck on beat 1 of the playing deck's phrase.

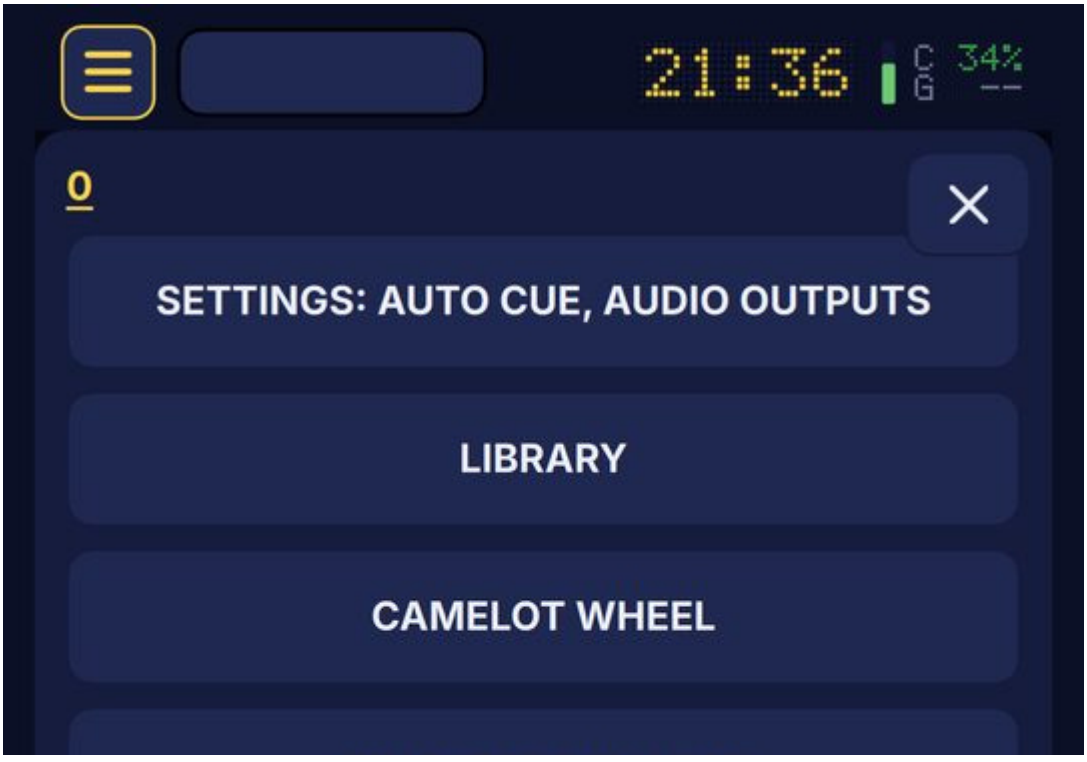
7. Keys

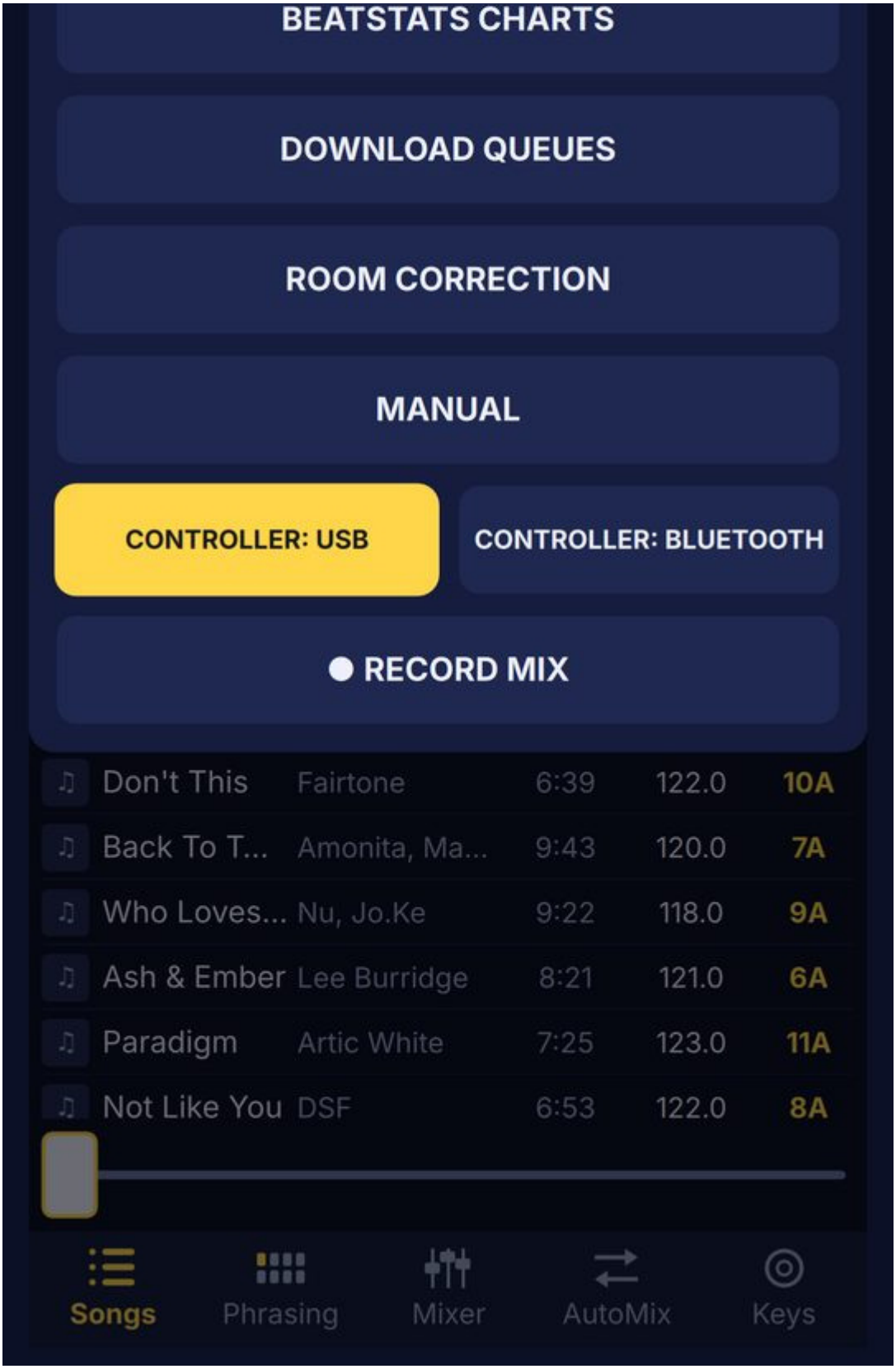




- The **Camelot wheel**: each deck's key is lit; neighbours on the wheel mix well.
- **KEY MATCH** shifts a deck to the matching key; off, it goes back to its own key. **– / +** move the key a semitone.

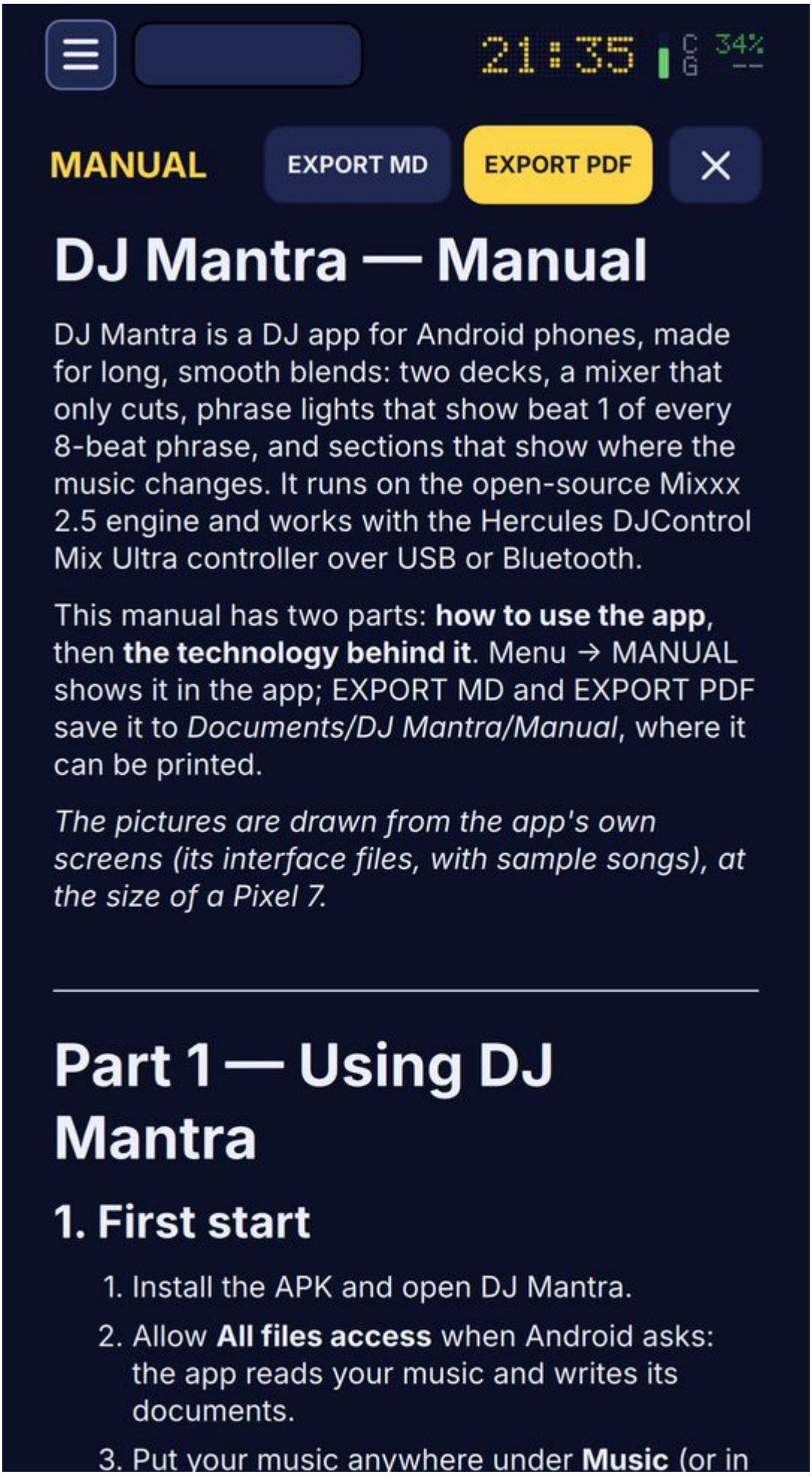
8. The menu





- **SETTINGS** (auto cue, audio outputs, waveform style, tempo range, controller, TIDAL login, recording ...).
- **LIBRARY**, **CAMELOT WHEEL**: the Songs and Keys tabs.
- **BEATSTATS CHARTS** (section 10), **DOWNLOAD QUEUES** (section 10), **ROOM CORRECTION** (section 11).
- **MANUAL**: this manual. **EXPORT MD** saves it with its pictures, **EXPORT PDF** as a printable PDF (A4), both to *Documents/DJ Mantra/Manual*.
- **CONTROLLER: USB / BLUETOOTH**: one link at a time.

- **RECORD MIX**: records what plays, lossless (FLAC or WAV), to *Music/djmantra_app/Recordings*.

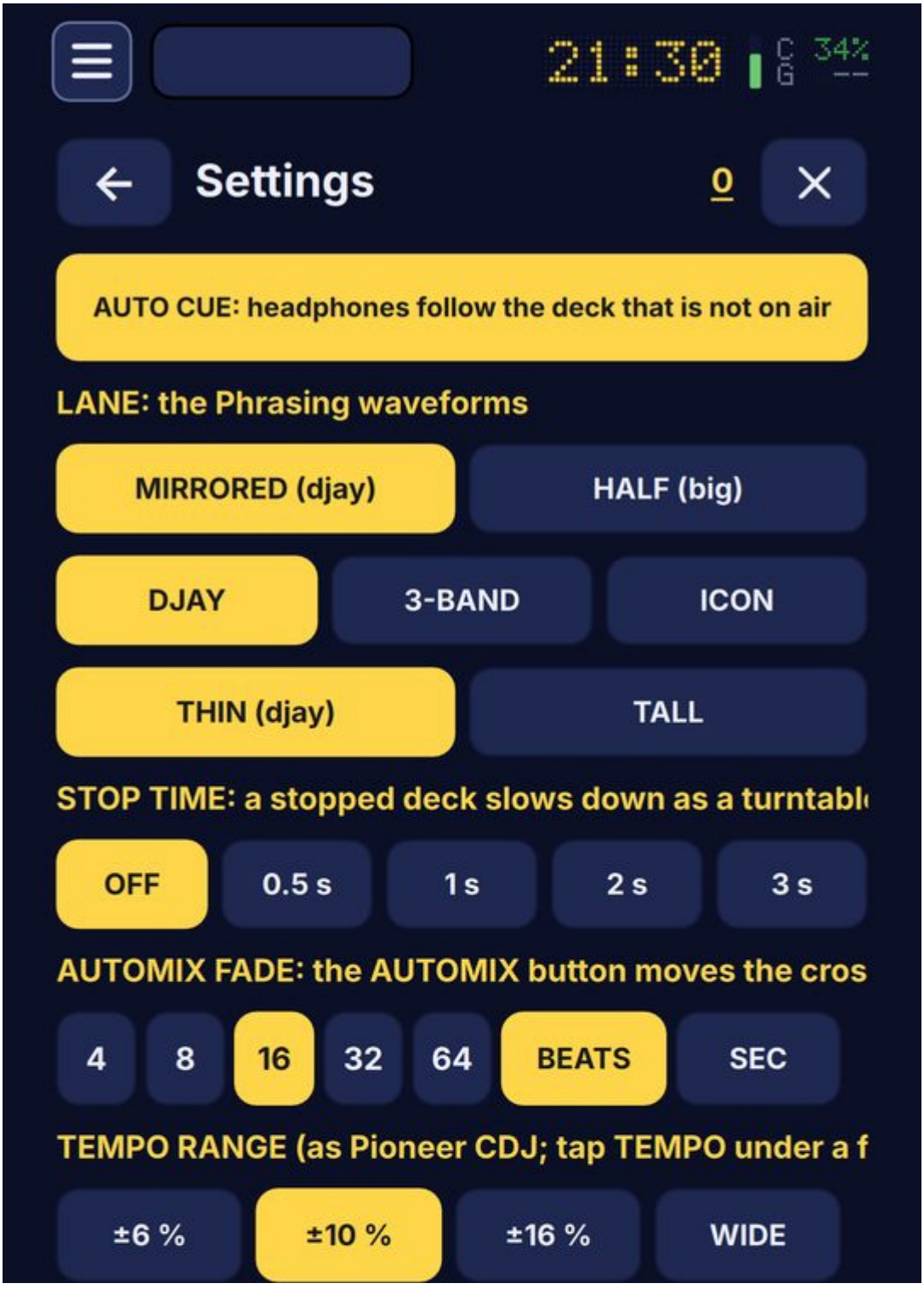


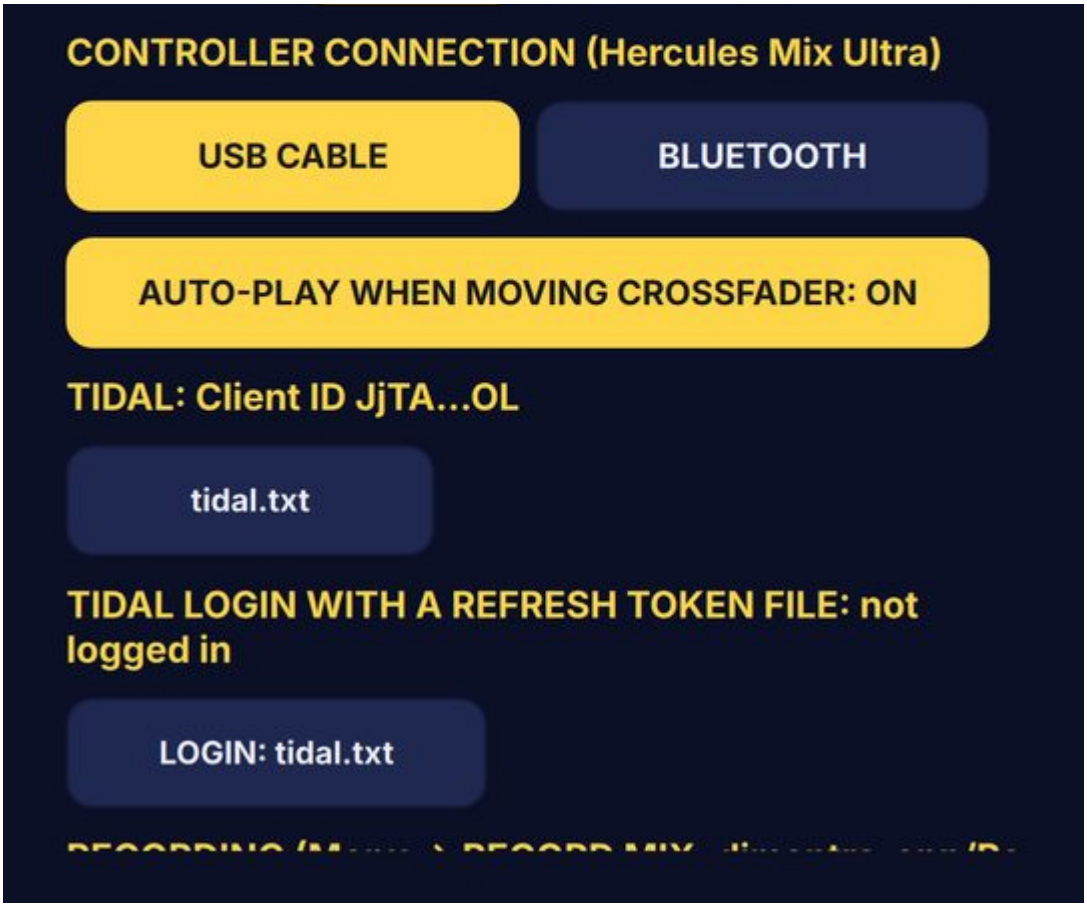
Music/djmantra_app). The app analyses every song once: tempo (BPM), beat grid, key, sections and loudness.

4. Connect the controller (optional): Menu → **CONTROLLER: USB** or **CONTROLLER: BLUETOOTH**.

2. The screen

9. Settings

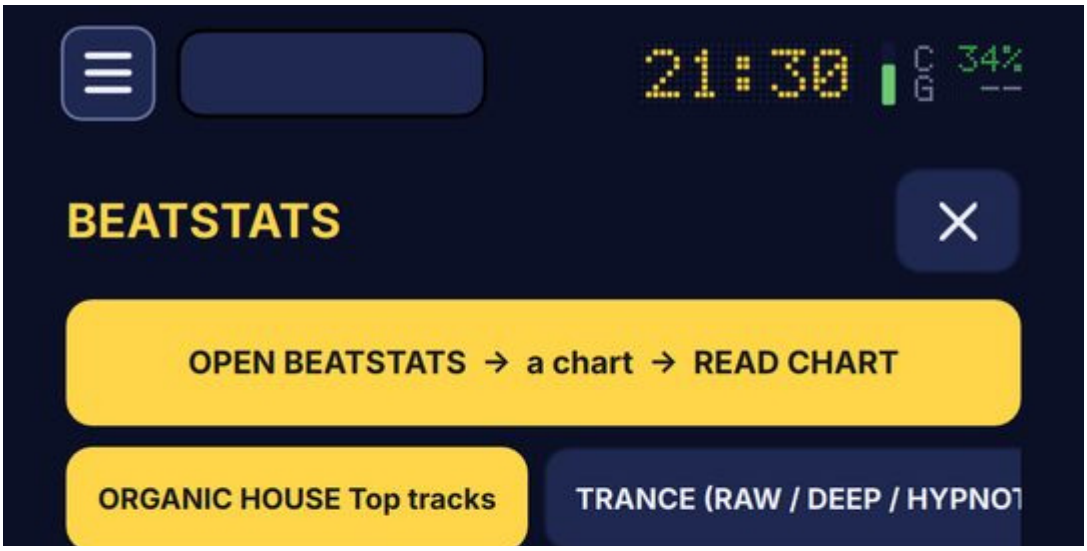




The most used:

- **AUTO CUE**: the headphones follow the deck that is not on air.
- **LANE**: the Phrasing waveforms mirrored (djay) or half (big); the **waveform style** DJAY (djay's colours), **3-BAND** (bass red, mids green, highs blue as layers, so sections show in the shape) or **ICON** (one colour a bar); THIN or TALL lanes.
- **STOP TIME**: a stopped deck slows down like a turntable (off, 0.5–3 s).
- **AUTOMIX FADE** in beats (4–64) or seconds.
- **TEMPO RANGE** ±6 / ±10 / ±16 % / WIDE, like a Pioneer CDJ.
- **CONTROLLER CONNECTION**, **AUTO-PLAY WHEN MOVING CROSSFADER**.
- **TIDAL**: the Client ID key file, and **LOGIN WITH A REFRESH TOKEN FILE** (section 10).
- **RECORDING**: format and file name.
- **CQ MIXER**: the delay of a connected CQ mixer in time with the music.

10. BeatStats charts, TIDAL and the download queues





Read a chart

1. Menu → **BEATSTATS CHARTS** → **OPEN BEATSTATS**. beatstats.com opens in the app's browser. Pass its "Verify you are human" check once (it is remembered).

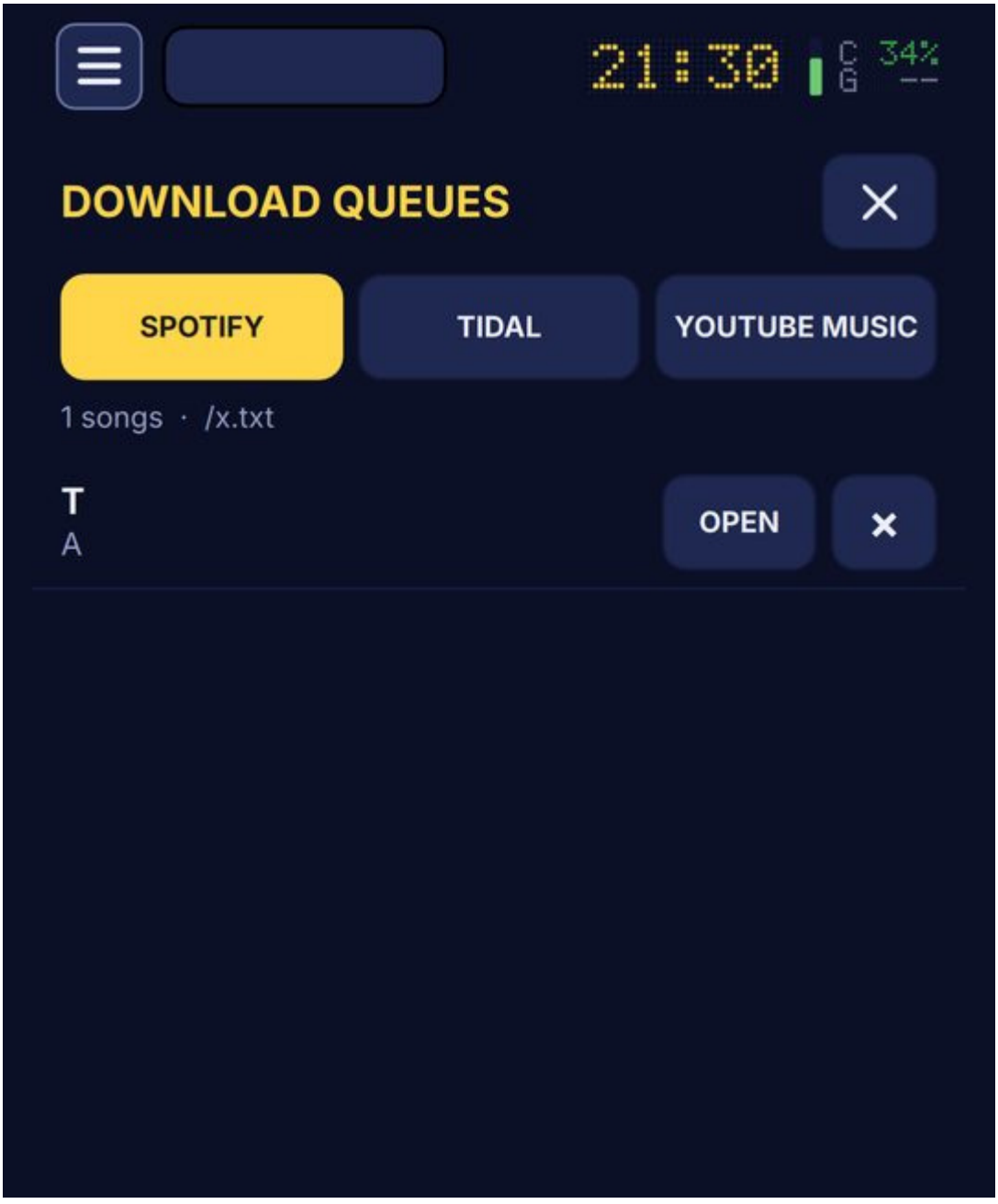
2. Open a chart (a genre's TRACKS list) and tap **READ CHART**. The app reads rank, title, artist and the Beatport ID of every song, then each song's ISRC (the code of the exact recording).
3. **CAPTURE** saves the whole page as one tall JPEG.

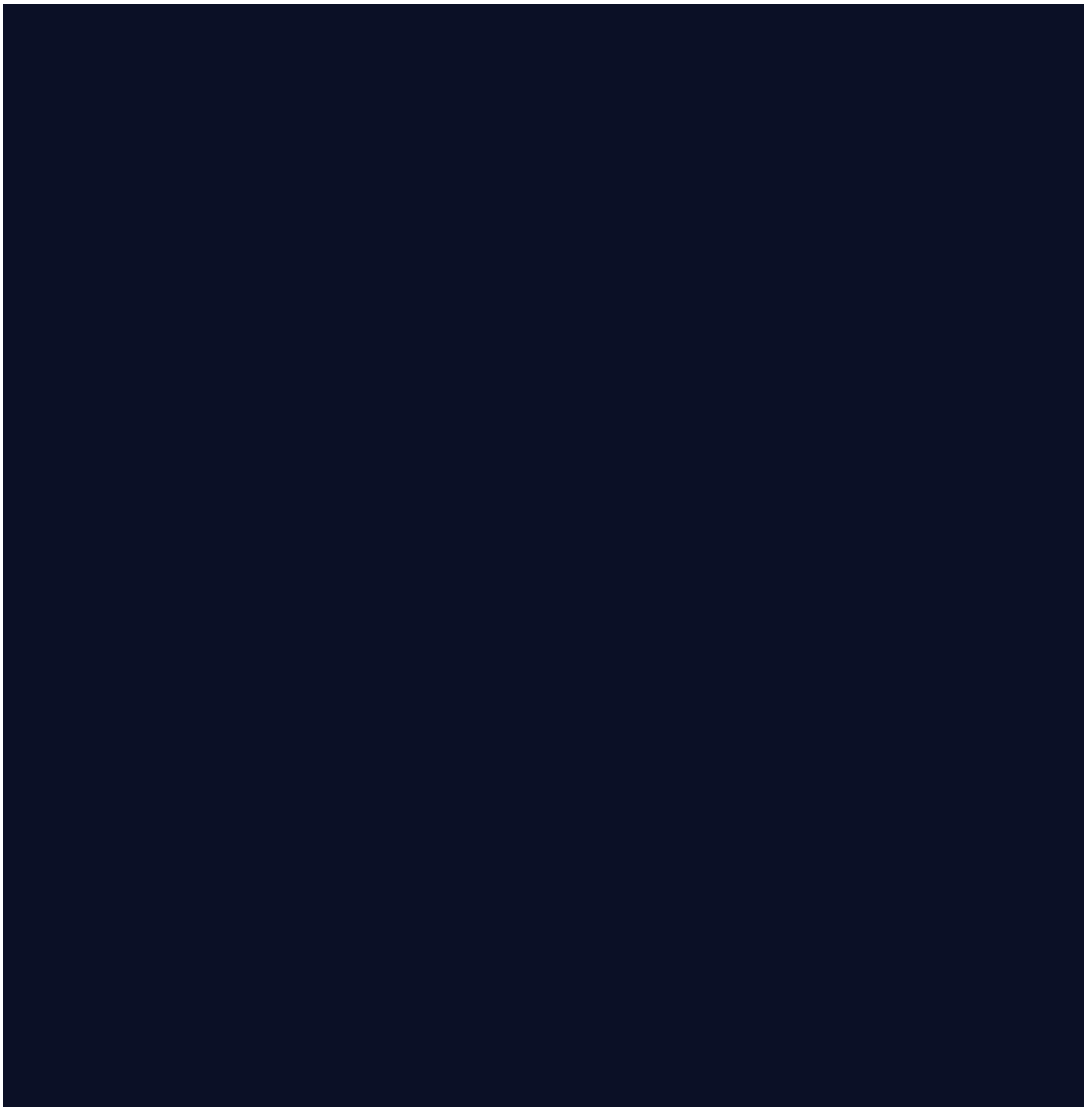
From a chart

- **SPOT / TIDAL / YT M**: open the song in Spotify, TIDAL or YouTube Music.
- **+**: add one song to a download queue; **+ ALL**: the whole chart at once.
- **TIDAL PLAYLIST**: the whole round in one tap:
    1. the chart becomes the TIDAL playlist **"BeatStats – *chart name*"** (unlisted). Each song is found by its ISRC, else by a careful search: the same song in another version if the named remix is not on TIDAL, never another song;
    2. the chart's songs that are **on your phone** go into the app playlist of the same name, ready for the decks;
    3. the others go into the **TIDAL download queue**.

TIDAL login. Settings → TIDAL. Either the browser login, or put a text file with your TIDAL **refresh token** (a line `TIDAL_REFRESH_TOKEN=...`, or the token alone) into *Documents/DJ Mantra/Keys* and tap **LOGIN:** followed by the file's name. The token is kept by the app and never shown.

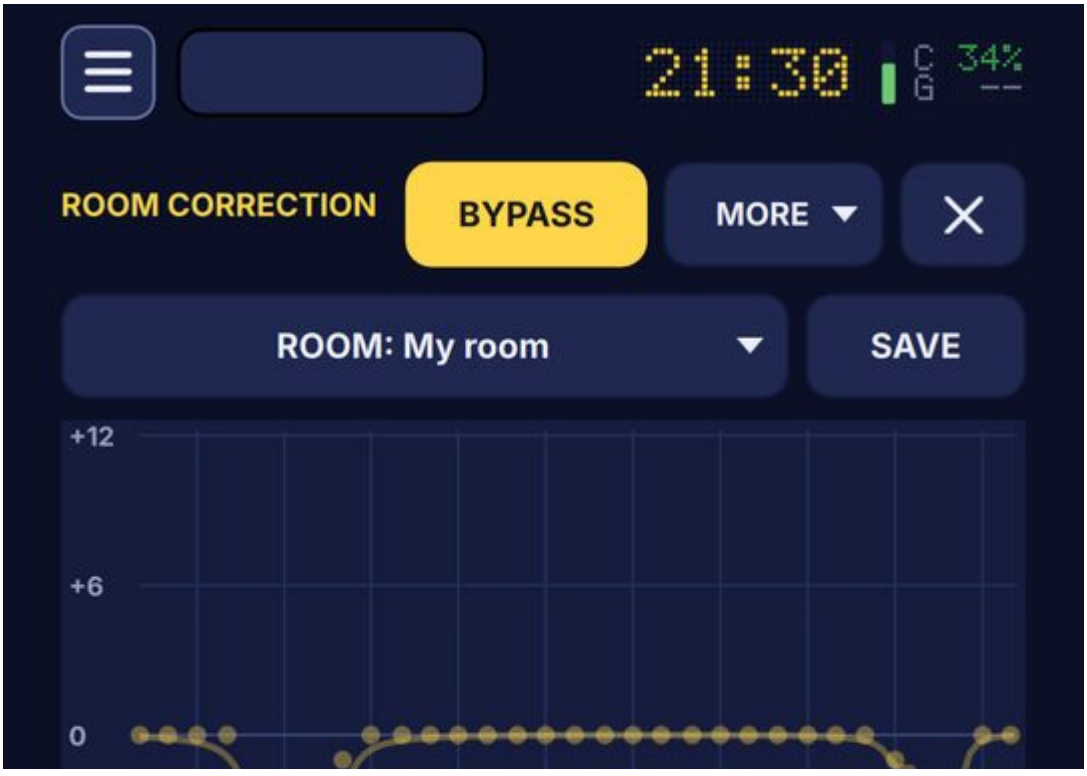
TIDAL plays 30-second previews in other apps. TIDAL gives DJ Mantra only previews, so songs play in the decks from your phone's files. Full TIDAL streaming in a DJ app needs TIDAL's DJ partnership.

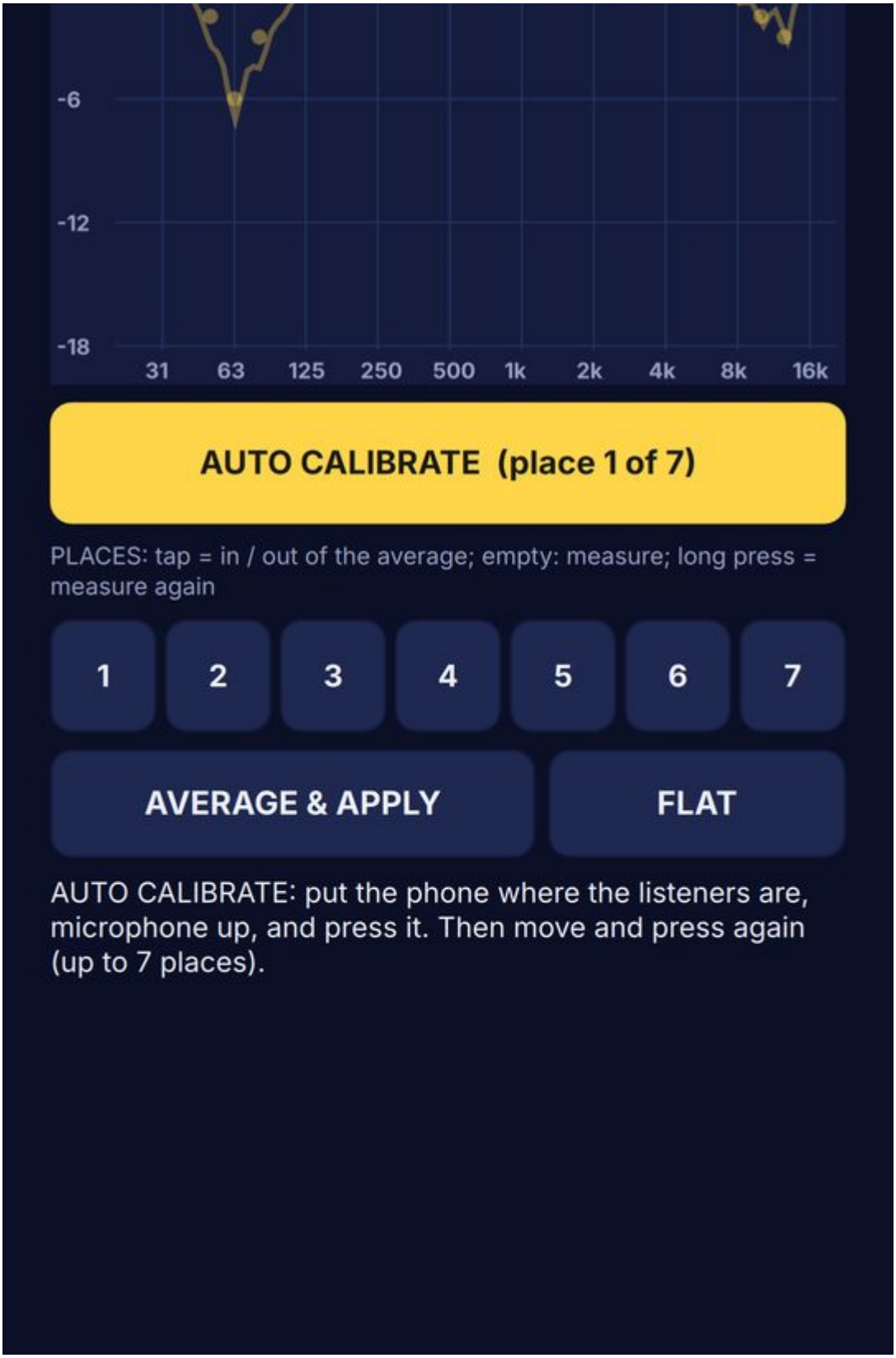




Download queues: one text file per service (Spotify, TIDAL, YouTube Music), one line a song: *Artist* | *Title* | *link*. Open a link to buy or download the song.

11. Room correction





Corrects the speakers for the room, like IK Multimedia's ARC.

1. Connect the speakers. Choose the microphone in MORE (the phone's, or a USB measurement microphone).
2. Put the phone where the listeners will be and tap **AUTO CALIBRATE**: the speakers play a sweep, the phone listens.
3. Move and repeat, up to **7 places** (tap a place to include it or not; long press measures it again).
4. **AVERAGE & APPLY**: the correction goes onto the main output (not onto the recording). **BYPASS** compares; **FLAT** clears it.
5. **SAVE** keeps the room (several rooms can be saved, chosen at the top).

MORE also holds the target (FLAT, HOUSE, CLUB), the strength (50 / 75 / 100 %), the test signal and its level, a live analyser and a short guide.

12. The Hercules DJControl Mix Ultra

Control	Does
PLAY, CUE, SYNC	as on screen; SHIFT + PLAY = stutter, SHIFT + CUE = back to the start, SHIFT + SYNC = key lock
LOAD	loads the highlighted song; SHIFT + LOAD = eject
BROWSE turn / press	opens the song list, moves, loads
Jog wheel	touch the top to scratch, turn the ring to nudge; SHIFT + top = fast seek
HIGH / MID / LOW	the EQ (or Neural Mix in that mode); with SHIFT the other mode; with pickup and ghosts
FILTER	the filter, with pickup; **SHIFT + FILTER = the depth of the deck's effect** (a pop-up shows its fader)
Pad mode buttons	HOT CUE = Songs, LOOP = Phrasing, FX = Mixer, NEURAL MIX = AutoMix, SHIFT + NEURAL MIX = Keys
SHIFT + FX (FX pads)	pads 1–8 = the 8 effects on that deck: tap = on / off, hold = only while held; SHIFT + pad = effects 9–16
NEURAL MIX pads	1–3 kill low / mid / high, 4 filter reset, 5–8 jump to beats 5–8; SHIFT + pad 5 = "1 is here"
LOOP pads	beat loops ¼ – 32 beats; SHIFT = loop rolls
SAMPLER pads	play samplers 1–8; SHIFT = stop

In every pad mode the pads light the current beat of the 8-beat phrase (the FX pads light the effect that is on).

13. Where the files are

What	Where
Your music	*Music* (and *Music/djmantra_app*)
Playlists	*Music/djmantra_app/Playlists*
Suno songs	*Music/djmantra_app/Suno*
Recordings	*Music/djmantra_app/Recordings*
BeatStats charts, captures, saved pages	*Documents/DJ Mantra/BeatStats*
Download queues	*Documents/DJ Mantra/Download_Queue*
Rooms	*Documents/DJ Mantra/Room_Corrections*
Key and token files	*Documents/DJ Mantra/Keys* (or Download, Documents)
This manual, exported	*Documents/DJ Mantra/Manual*

Every time the app saves a file, a message says its name and folder.

Part 2 — The technology

14. The engine

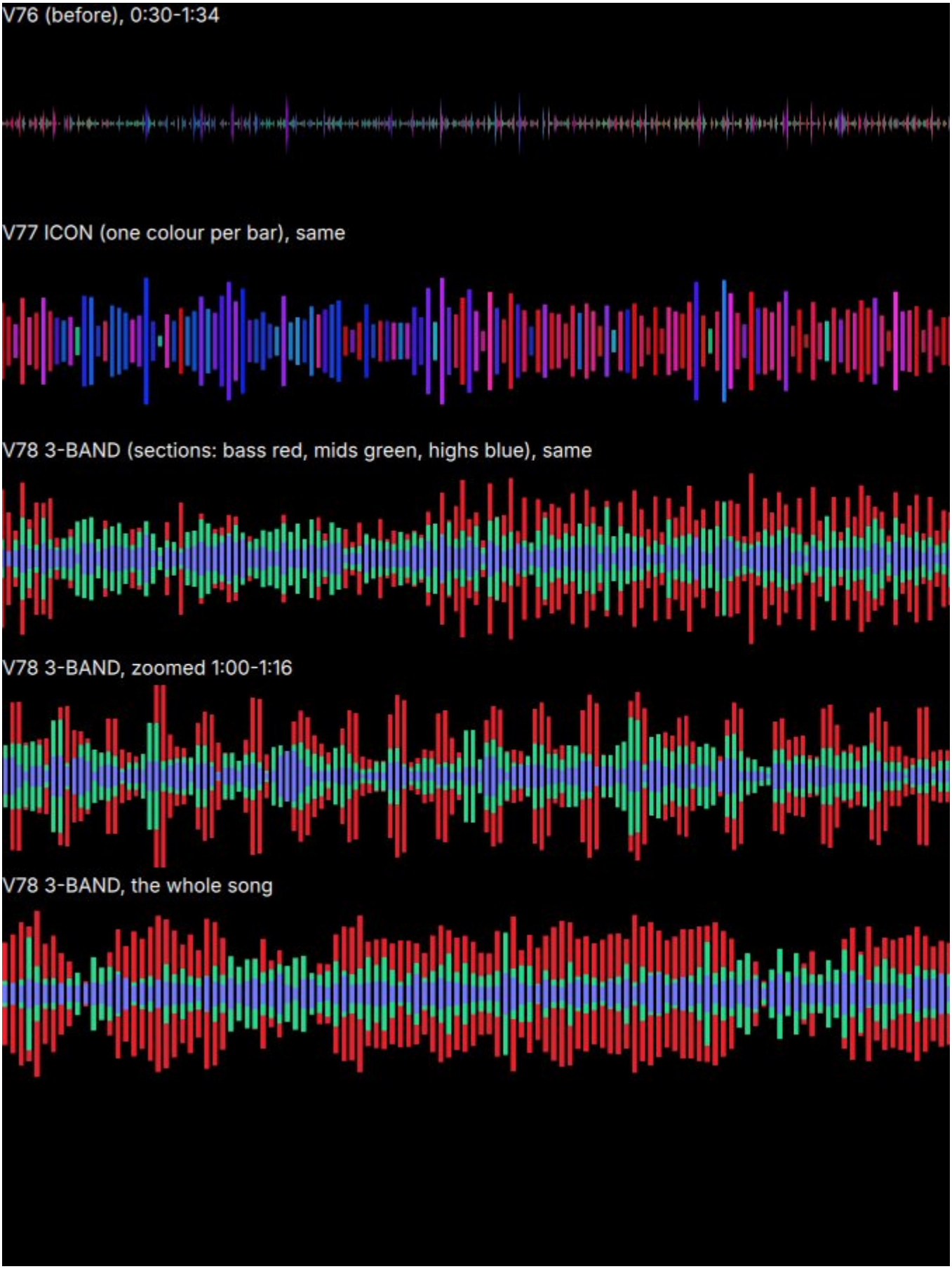
DJ Mantra is built on **Mixxx 2.5**, the open-source DJ software (GPL), ported to Android: the same audio engine (decks, mixer, effects, time-stretching, sync), its library and analysis, compiled for the phone's ARM processor with the Android NDK and Qt 6. The interface is new, written in **Qt Quick** (QML) for a phone held upright; Mixxx's desktop screens are not used. Sound goes out through Android's low-latency audio (Oboe / AAudio), to the phone, a USB sound card or a USB-C audio output, with the headphones on a second output when there is one.

15. Analysis

Each song is analysed once, in the background, and the results are kept in the library:

- **Beats and BPM**: a beat grid; the first beat of the grid is placed on the kick.
- **Key**: shown in Camelot notation (8A, 11B …) for harmonic mixing.
- **Sections**: where the arrangement changes (bass in or out, vocals in or out, a new pattern), found from changes in the sound along the beat grid and aligned to the 8-bar phrase grid. The app counts down to the next change; it does not name the sections. Settings has options for how sensitive the detection is.
- **Loudness**: every song is normalised so its peak sits at **−1 dBFS** (quiet songs raised, loud ones lowered, within ×¼ … ×8), so songs play equally loud and the faders set the volume. There is no GAIN knob.

16. The waveforms



The scrolling waveform is drawn by the graphics processor (a fragment shader) from the analysis: for every moment the energy in three bands, **bass** (below 600 Hz), **mids** and **highs** (above 4 kHz). The bars are solid and sharp to the device pixel, and they belong to the song, so they do not shimmer while scrolling. Three styles:

- **DJAY**: djay's mixed colours (the default).

- **3-BAND**: three nested layers per bar, bass red at the back, mids green, highs blue-violet in front. The colours never mix, so a breakdown (no red), a drop (wide red) or vocals (green) read as blocks: sections show in the shape itself.
- **ICON**: one pure colour per bar, as in the app's icon.

At deep zoom the real sound wave is drawn as a thin line.

17. The mixer

- **EQ and Neural Mix** are one effect of DJ Mantra's own (StemEQ), working on a short-time spectrum (FFT of 1024 points, about 23 ms): one mask per frequency, the masks adding up to the original sound at 0 dB.
    - **EQ**: three bands split at the mixer's EQ frequencies, cut only: the whole travel of the knob (127 steps on the controller) is spent on cutting, fine near 0 dB.
    - **Neural Mix**: **drums** = what is short in time (harmonic / percussive separation), **vocals** = what is steady and in the middle of the stereo image (150 Hz – 7 kHz), **harmonies** = the rest. It is light signal processing that runs on any phone, not a neural network, so the separation is approximate.
    - Both sets act together; the two rings of each concentric knob are the two sets.
- **FILTER**: modelled on the Pioneer DJM filter (resonant low-pass and high-pass, off in the middle).
- **Effects**: Mixxx's beat-synced effects with DJM-like settings, plus DJ Mantra's UNDERWATER, COOLDOWN and ECHO OUT.

18. Room correction

The speakers play a **logarithmic sine sweep**; the phone records it, and the room's response is found by dividing out the sweep (Farina's method), gated to the first 0.68 s so late reflections do not count. It is read in **31 third-octave bands**. The places are averaged by power, compared with the target curve relative to 200 Hz – 2 kHz, limited (no correction below the speakers' lowest usable frequency or above 16 kHz, at most −12 / +6 dB), smoothed, shifted so the correction only cuts, and applied at the chosen strength through a 31-band parametric equaliser on the main output. A quiet or implausible measurement is refused rather than applied.

19. The controller

The Mix Ultra speaks MIDI over USB or Bluetooth LE. Its mapping is a script: the knobs are read in 14-bit resolution; EQ, FILTER and the effect depth use **pickup** (a knob takes over only when it reaches the value, as in Ableton Live), and the app shows the **ghost** of a knob that has not picked up yet. The pad lights show the phrase (one pad per beat).

20. TIDAL, BeatStats and the playlists

- **TIDAL**: the official TIDAL API (OAuth 2.1 with PKCE, no secret in the app). Songs are found by **ISRC** first; without one, a ladder of searches from exact to loose; a match needs the same title and one of the artists, prefers the named version, then the original, and never takes another song, a cover or a karaoke version. Requests wait when TIDAL says "too many requests". Playlists are created unlisted, and sending a list again never adds a song twice.
- **BeatStats**: read from the page in the app's own browser (no OCR, no AI): the exact rows, then each song's ISRC from its Beatport page.
- The same matching is used by the *Playlist Generator* that makes TIDAL playlists from reference songs; both are described in the Mantra Manifest (*modules/tidal-playlists.md*).

21. Limits

- TIDAL gives this app **30-second previews** only; full songs play from the phone's files.
- beatstats.com and Beatport protect themselves with Cloudflare: the first visit needs a person to pass the check.
- Neural Mix separates drums, vocals and harmonies by signal processing, not by a neural network: the separation is approximate.

DJ Mantra — built on Mixxx (GPL). This manual describes version V86.